WORLDSEED

Developer-Ready Design & Implementation Blueprint

RC7 Update - Empty Hands Early-Game Overhaul

Defining rule
Model causes and rules; let outcomes emerge. RC7 applies that principle to the opening game: the character begins with nothing, and primitive technology, food, shelter and industry emerge from resources that physically exist in the generated world.

Status: RC7 implementation and engineering validation complete. Final archive verification is the release gate performed after document generation.

Engine: 1.0.0-rc7 | Save schema: 3 | Worldgen: 3 | Natural-resource generation: 2

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1. Executive Summary

WORLDSEED is a terminal-first, persistent fantasy-world simulation in which the world - not a single avatar - is the durable protagonist. Deterministic terrain provides effectively unbounded space; simulation fidelity decreases with distance; only changed or realized state is persisted; ecology, settlements, economies, history, knowledge, religion and magic interact through common events, entities and state.

RC7 focuses on a weakness identified through actual play: WORLDSEED was already broad and complex, but its opening hours skipped too much of the physical survival sandbox. RC7 rebuilds that opening around empty hands, material bootstrapping, fieldcraft, useful furniture, hunting/carcass processing, dynamic cooking/nutrition, crop soil roles and richer habitat-aware ecology.

RC7 release rule
A new character receives no supplies. Fairness comes from a viable landscape, not a starter kit or scripted objective.

1.1 Verified RC7 scope

Area	Verified RC7 result
Starting state	Inventory is empty; point-buy character capability only.
Content	105 items; 52 recipes; 23 structures; 8 crops; 22 natural resources; 17 authored animals; 4 cooking methods.
Regression	156 automated tests passed.
Bootstrap	10/10 independent generated worlds reached fire, timber, carcass processing, cooked food and diet history from zero supplies.
Start balance	20/20 sampled starts were empty and primitive-bootstrap capable; no weak starts.
Performance	~15.11 ms/chunk; ~1.62 ms movement; ~3.97 ms movement + 100x30 render.
Long worlds	3 x 25-year and 1 x 100-year FULL simulations completed with zero invariant errors.

2. Product Principles

Principle	Implementation meaning
Systemic causality	Model resource, need, process and response; do not script the desired story.
Persistent consequences	Player and simulation actions alter the same durable world.
Deterministic potential	Unvisited infinity remains generated potential until required.
Layered fidelity	Individuals locally; cohorts/ledgers/events at distance.
Compositional content	Materials, tool qualities, processes, components and data definitions compose into outcomes.
Discoverability	The engine can know more than the character; UI reveals what the character has learned.
No clutter for realism alone	An item or furniture piece needs a mechanical consumer or meaningful downstream process.
Terminal native	Use information density and navigation hierarchy; do not imitate a graphical UI badly.

3. Architecture Snapshot

3.1 Simulation fidelity

Tier	State	Fidelity
A	Immediate player area	Exact individuals, movement, combat/anatomy, fluids/fire, local pathing.
B	Nearby	Persistent individuals at reduced decision cadence.
C	Regional	Population cohorts, settlement ledgers, trade, migration, outbreaks.
D	Distant realized	Coarse daily/monthly equations and scheduled events.
E	Unrealized	Deterministic potential only; no persistent population explosion.

3.2 Persistent-state rule

Untouched terrain is reconstructed from seed + generator version + coordinate. Saves primarily store deltas, instantiated entities, discoveries and aggregate living-world state. The same principle applies to RC7 natural resources: deterministic placement is versioned separately, while harvested/regrowing features persist only the changes required to reconstruct their current stage.

3.3 Command authority

The UI requests commands; authoritative simulation validates resources, tools, time, collision and outcomes. Rendering remains observational. RC7 cooking, carcass work, ranged hunting, farming, preservation and construction use the same command/state boundary rather than mutating inventory directly from menus.

4. RC7 Zero-Supply Opening

4.1 Empty-hands contract

- Inventory is exactly empty at new-character creation.
- No money, food, tools, medical supplies, seeds, container or hidden origin kit is granted.
- Character point-buy affects effectiveness, not initial possessions or prescribed story.
- Spawn scoring can reject unfairly boxed or nonviable starts, but cannot disclose where resources are.

4.2 Primitive dependency graph

The intended first technology graph is:

Bootstrap
Loose stone + fallen branches + plant fiber/bark -> flakes + cordage -> knife/axe/pick/hammer/digging stick -> fire/shelter/work surface -> timber/mining/hunting/farming -> preserved food/leather/better stations -> advanced production.

4.3 Start fairness without handholding

The final 20-seed start audit found zero weak starts. Within radius 14 the median sampled feature counts were approximately: stone 100.5 cells, branch 36, fiber 39, edible forage 45.5, water 123 and trees 206. The worst nearest observed distance was stone 13, branch 9, fiber 6, edible 5, water 10, tree 3. These numbers are generator diagnostics only; the player is not shown them.

5. Item and Craft Economy

5.1 Tool-quality model

Processes request capabilities such as cut, skin, butcher, scrape, chop, mine, hammer, till, dig and sew. A crude or advanced tool can provide the same quality at different strength/speed. This avoids recipe logic that depends on one exact item ID and supports substitution and material progression.

5.2 Anti-clutter contract

An item is accepted only when it is directly usable or has a downstream consumer: process input, construction material, equipment component, storage/fuel/ammunition, agriculture, trade backed by utility, or a knowledge/cultural artifact with consequences. The automated content audit rejects undefined references, missing prose, functionless furniture and obvious orphan content.

5.3 Primitive recipe design

Input opportunity	First outputs	What it unlocks
Loose stone	flakes, hammer/pick heads, sling stones	cutting, mining, striking, ranged ammo
Fallen branches	hafts, fuel, digging stick, spear parts	bound tools, fire, shelter/furniture
Brush/bark/fiber	cordage/bindings	tool assembly, sling, structures
Tree + chopping quality	wood, bark	larger construction, workstations, fuel
Rock/ore + mining quality	stone/ore	masonry, metallurgy

6. Hunting, Combat and Carcass Processing

Authored animals carry mass, habitat/climate envelope, health, aggression, group data, named attacks and carcass profiles. Procedural species use the same runtime contracts. Local individuals remain linked to regional population cohorts so local kills affect aggregate ecology.

6.1 Carcass lifecycle

Step	Mechanical effect
Kill	Creates persistent carcass; combat damage contributes to carcass quality loss.
Skin	Produces fresh hide; best performed before butchering.
Butcher	Produces meat, fat and offal; butchering first damages hide yield.
Debone	After butchery, recovers bones and sinew and completes carcass processing.
Delay	Time reduces usable quality; neglected carcasses become unsafe.

6.2 Product sinks

Product	Current/future sinks
Meat	roast, stew, broth, drying, trade
Hide	scrape -> rawhide or tanning -> leather
Bone	tools, needles, broth, fertilizer paths
Sinew	strong cordage/equipment
Fat	cooking, soap, preservation
Offal	food/bait; future feed/fertilizer extensions

7. Cooking, Preservation and Nutrition

7.1 Ingredient-composed meals

Current methods are roast, stew, porridge and broth. Methods define station/cookware/water constraints and ingredient roles; the resulting meal derives nutrition from the actual selected ingredients instead of enumerating every combination as a fixed recipe.

7.2 Nutrition model

Layer	Tracked values
Immediate	satiety/hunger and hydration
Meal composition	calories, protein, carbohydrates, fat, fiber
Diet variety	protein, vegetables, grains, fruit, fats
Rolling history	seven-day totals and recent meals
Effects	bounded short-term recovery, stamina or cold-resilience bonuses

7.3 Preservation

Prepared meals have finite shelf life. Carcasses decay. Drying and tanning are timed processing jobs tied to physical stations. Preservation therefore converts perishable opportunity into durable stored value rather than simply creating alternate item names.

8. Farming, Forage and Ecology

8.1 Crops

Crop set	Distinct data
Wheat, barley, beans, turnip, cabbage, carrot, onion, flax	growth duration, temperature/water envelope, N/P/K demand, disease family, seed/harvest outputs

Farming requires a tilling-quality tool. Nutrient use is crop-owned rather than hard-coded, so crop rotation can matter naturally. Wild plant resources can produce seed where appropriate; farming can therefore emerge from exploration rather than a starter packet.

8.2 Natural resources

The 22 authored natural-resource definitions include multiple trees, berries, edible greens/root plants, herbs, fungi, wetland plants, wild grain, clay, branches, loose stone and boulders. Placement uses terrain, climate, season, density, patch scale and deterministic local variation. Structural features such as trees retain identity through stump/sapling/mature stages.

8.3 Fauna

Regional abundance samples actual habitat suitability; local animals materialize on suitable passable cells. The player does not receive exact population statistics without sufficient Naturalism/Survival or reliable reports.

9. Functional Construction and Furniture

Piece	Purpose
Campfire/hearth	warmth, cooking, light, primitive heat processes
Brush lean-to	shelter/exposure protection
Sleeping mat/bed	rest/recovery

Piece	Purpose
Storage basket/chest	physical storage
Crude work surface/workbench	crafting efficiency
Butcher table	carcass-processing efficiency
Drying rack	hide/meat/herb drying
Tanning frame	leather production
Forge	metallurgy
Walls/gates/roofs/supports	enclosure, protection, structural shelter
Bridge	crossing

Purely decorative furniture remains deferred until beauty, prestige, culture or social-space systems can make decoration mechanically meaningful.

10. Terminal UX for Complex Crafting

The interface is purpose-grouped instead of flat-list driven. Crafting and construction entries show READY/NEEDS before selection. Detail pages use readable material names, station/tool/skill requirements, time, description and the role enabled by the result. Cooking offers a live nutrition preview. Carcass menus show only valid next operations.

- W/S or arrows navigate; D/Enter selects; A/Esc goes back one layer.
- The UI describes player-facing consequences, not simulation internals.
- Long records use pagers; menu selection/scroll state is retained by the UI navigation stack.

11. Research Basis

RC7 research was used as architectural inspiration rather than copied content.

Reference	Useful principle adopted
Cataclysm: Dark Days Ahead JSON documentation	Data-driven items/recipes/requirements and reusable tool qualities instead of exact-tool conditionals.
Vintage Story farming documentation	Distinct soil nutrients and crop rotation as a meaningful agricultural constraint.
Vintage Story meal/nutrition documentation	Ingredient-composed meals and separation of immediate food value from longer-term dietary variety.
UnReal World hideworking documentation/community references	Distinct skinning/butchering/hide-processing stages and quality consequences.
Textual/Ratatui terminal UI patterns	Retained navigation state, explicit screens/modes and model-update-view separation.

11.1 Reference URLs

- https://docs.cataclysmdda.org/JSON/JSON_INFO.html
- https://docs.cataclysmdda.org/JSON/ITEM.html
- https://wiki.vintagestory.at/Farming
- https://wiki.vintagestory.at/Meal
- https://wiki.vintagestory.at/Food
- https://www.unrealworld.fi/wiki/index.php?title=Hideworking
- https://textual.textualize.io/guide/screens/

12. Validation and Release Gates

Gate	Final RC7 status
Compilation	PASS
Automated tests	156/156 PASS
Content audit	PASS
Zero-item bootstrap	10/10 seeds PASS
Start balance	20/20 empty and bootstrap-capable; 0 weak starts
Replay	deterministic; RC7 hash f65ea5583575a883f45195a78c6f86d7c5ed64dc
Worldgen benchmark	~15.11 ms/chunk
Movement benchmark	~1.62 ms/step; ~3.97 ms incl. full render
25-year soaks	3/3 seeds; zero invariant errors
100-year soak	zero invariant errors; ~1.96 MB state

12.1 Invariant focus

- No negative population/stock/wealth counts.
- No orphan settlement/NPC/family/office/infection references.
- No unbounded trade-route complete graph.
- Unrealized infinity remains deterministic potential.
- Content references resolve; items/furniture have utility.
- Same seed + same commands remains deterministic.

13. Completed vs Remaining

System	Status
World/determinism/persistence	Complete foundation
Terminal UI/input/navigation	Mature, continuing polish
Exploration/discovery	Implemented
Primitive early game	RC7 complete
Hunting/carcass economy	RC7 complete
Dynamic cooking/nutrition	RC7 complete
Farming/soil	Substantial, expandable
Ecology/procedural fauna	Substantial
Construction/rooms/roofs	Substantial
Combat/anatomy/equipment	Substantial
NPC society/economy/factions	Substantial
History/knowledge/religion/magic	Substantial
LOD/offline simulation	Implemented/hardened
Epidemiology	Implemented

13.1 Post-RC7 depth priorities

- Trapping, fishing and physical tracking/sign systems.
- Smoking, salting, fermentation, pickling and cold storage.
- Livestock/domestication/breeding and deeper animal social/territorial behavior.
- Compost/manure, pests, irrigation networks and perennial crops.
- Recipe knowledge/proficiencies/experimentation instead of universal recipe knowledge.

- Advanced textiles/weaving/clothing construction.
- Deeper aesthetics/comfort/prestige so decorative furniture gains real social value.
- Search/favorites and keybinding customization for very large craft/build catalogues.
- Balance telemetry from real play sessions and additional biome/start sampling.

14. Future Development Rules

1. Never give a new character supplies merely to mask an unreachable resource graph.
2. Every new mechanic must interact with at least two existing systems before it is considered a completed feature.
3. Prefer tool qualities, material properties, components and process definitions to one-off item checks.
4. Keep player knowledge separate from simulation knowledge.
5. Keep distant simulation aggregated and event-driven.
6. Preserve deterministic RNG namespace isolation.
7. Run content audit and zero-item bootstrap tests after any primitive resource/recipe change.
8. Run long-world invariants after economy/ecology/AI/history changes.
9. Do not add decorative clutter until another system consumes its aesthetic/social value.

Appendix A - RC7 Content Counts

Content type	Count
Items	105
Recipes/process recipes	52
Structures/furniture	23
Crops	8
Natural resource definitions	22
Authored animal species	17 + procedural species
Dynamic cooking methods	4

Appendix B - Release Commands

python -m compileall -q worldseed

python tools/audit_content.py

python tools/bootstrap_smoke.py --seeds 10

python tools/balance_early_game.py --seeds 20

python tools/replay.py tests/fixtures/golden_replay.json

python tools/bench_worldgen.py

python tools/bench_movement.py

python tools/soak_matrix.py --years 25 --seeds 3 --policy full

python tools/soak_matrix.py --years 100 --seeds 1 --policy full